A
Mini-Project Report on

AI BASED QUIZ SYSTEM

Submitted in partial fulfillment of the requirements
for the degree of
BACHELOR OF ENGINEERING
IN
Computer Science & Engineering
Artificial Intelligence & Machine Learning

by

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Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY
(Approved by AICTE New Delhi & Govt. of Maharashtra, Affiliated to University of Mumbai)
(Religious Jain Minority)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

CERTIFICATE

This is to certify that the project entitled "**AI Based Quiz System"** is a bonafide work of Yash Penkar (22106118), Prarthana Patil (22106035), Rutuja Pawar (22106043), Dhruv Sawant (22106015) submitted to the University of Mumbai in partial fulfillment of the requirement for the award of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning).**

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(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Project Report Approval

This Mini project report entitled "**AI Based Quiz System"** by **Yash Penkar, Prarthana Patil, Rutuja Pawar and Dhruv Sawant** is approved for the degree of ***Bachelor of Engineering*** in ***Computer Science &Engineering***, (AI&ML) ***2024-25***.

External Examiner: _____

Internal Examiner: _____

Place: APSIT, Thane
Date:

Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission hasnot been taken when needed.

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ABSTRACT

The increasing reliance on digital education and online assessments has led to the need for efficient and scalable quiz management systems. Traditional paper-based quizzes are often cumbersome, time-consuming, and prone to manual errors. To address these challenges, this project presents a web-based quiz application that leverages modern web technologies to streamline the process of quiz administration, participation, and result tracking. The system is built using Flask, a lightweight web framework, and MySQL, a robust relational database management system. It offers secure user authentication, allowing individuals to register, log in, and attempt quizzes in a controlled environment. Users can engage with interactive quizzes. The platform also supports structured question management, ensuring that quizzes remain dynamic and well-organized. A key advantage of system is its scalability and adaptability. It is designed to handle multiple users simultaneously, making it suitable for educational institutions, training programs, and online learning platforms. Additionally, automated scoring and data analysis provide valuable insights into user performance, reducing the burden on educators and administrators. Future enhancements may include AI-driven question generation and personalized learning paths to further enhance user engagement and educational effectiveness.

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CHAPTER 1

INTRODUCTION

1.INTRODUCTION

In today's digital era, technology continues to revolutionize various aspects of life, including education and skill development. As the world becomes increasingly interconnected, the need for efficient and scalable learning solutions has never been more critical. With advancements in technology, online learning has become an essential part of modern education, transforming the way students, professionals, and organizations acquire knowledge and assess skills. The increasing demand for flexible, accessible, and scalable learning solutions has led to the widespread adoption of digital platforms that facilitate remote education and training. Traditional quiz-taking methods, such as paper-based exams, are often inefficient, time-consuming, and prone to errors, making them less suitable for large-scale learning environments. Moreover, manual grading and result compilation add to the administrative burden, slowing down the assessment process and limiting real-time feedback. To address these challenges, institutions and businesses are shifting toward automated, technology-driven assessment solutions that enhance efficiency, accuracy, and user experience. A well-designed online quiz system not only simplifies evaluations but also provides valuable insights into learners' performance, enabling educators and trainers to make data-driven decisions. By leveraging modern web technologies, secure authentication mechanisms, and intelligent automation, online quiz platforms have become an indispensable tool for education, corporate training, and skill development programs.

CHAPTER 2

LITERATURE SURVEY

1. LITERATURE SURVEY

2.1-HISTORY

The concept of quizzes and assessments has been an integral part of education for centuries, evolving significantly with technological advancements. Traditional assessments were conducted through oral examinations, written tests, and paper-based quizzes, which required manual grading and extensive administrative effort. These methods, although effective in evaluating knowledge, were often time-consuming and lacked scalability, making large-scale assessments challenging. With the advent of computers and the internet, digital assessments began to emerge in the late 20th century. Early computer-based testing systems were limited to simple multiple-choice questions and required dedicated software installed on local machines. The rise of web technologies in the early 2000s revolutionized online learning, enabling the development of Learning Management Systems (LMS) that integrated quiz functionalities. Platforms like Moodle, Blackboard, and Google Classroom facilitated automated grading, instant feedback, and structured question banks, enhancing the efficiency of assessments. In the early 2010s, the rapid development of mobile technologies further expanded the reach of quiz systems. The rise of smartphones and tablet devices made quizzes more portable and accessible, allowing learners to engage with assessments anytime and anywhere. This mobile-first approach changed the dynamics of digital learning, providing greater flexibility and convenience to users. The demand for scalable, flexible, and intelligent assessment systems led to the development of AI-powered quiz platforms in recent years. These systems incorporate machine learning and natural language processing to generate dynamic questions, provide adaptive learning experiences, and analyse learner performance in real-time. Gamification principles also gained traction during this period, with platforms integrating badges, leaderboards, and rewards to boost user engagement and motivation. This approach became particularly popular in corporate training and online education platforms, enhancing the learning experience while maintaining a competitive edge. Around the same time, the use of big data and analytics in educational platforms provided new insights into learner behaviour, helping educators refine their teaching strategies. By analysing performance trends, quiz systems could offer personalized feedback and suggest tailored learning paths, fostering a more individualized approach to education. With advancements in cloud computing, quizzes have become more accessible, allowing users to participate in assessments remotely from any device. More recently, the rise of artificial intelligence and deep learning models has enabled quiz systems to become even more intelligent, with capabilities such as natural language

understanding and automatic question generation. These systems are increasingly able to adapt to the learner's skill level, offering personalized challenges and feedback, making learning more dynamic and effective. Today, web-based quiz applications play a crucial role in education, corporate training, and professional certification programs. They offer secure authentication, structured question categorization, and automated evaluation, making assessments more efficient, reliable, and engaging. The continuous evolution of technology is further enhancing the capabilities of online quiz systems, shaping the future of digital learning and assessments.

2.2-LITERATURE REVIEW

[1] "AI-BASED QUIZ SYSTEM FOR PERSONALISED LEARNING", Xia Wang, Silke Wrede, Lars van Rijn, Joachim Wöhrle in 16th annual International Conference of Education, Research and Innovation, November 2024.

The AI-Based Quiz System for Personalized Learning in Wang_AI_QuizSystem2023.pdf uses various modules, libraries, and packages for efficient quiz generation and adaptive learning. The backend relies on Python, integrating Neural Networks (NNs) and Natural Language Processing (NLP) with tools like BERT, RoBERTa, and OCR for question extraction. A Knowledge Graph Manager (KGM) is used to track learners' progress. The system integrates with Learning Management Systems (LMS) such as Moodle via a quiz plug-in. Additionally, machine learning (ML) models help in generating personalized quizzes based on learners' knowledge levels.

[2] "Automatic generation of multiple choice questions for e-assessment": A. Santhanavijayan*, S.R. Balasundaram, S. Hari Narayanan, S. Vinod Kumar and V. Vignesh Prasad in International Journal of Signal and Imaging Systems Engineering · January 2017.

The research found that frequent use of MCQs in assessments can sometimes encourage surface-level learning, where students focus on memorizing answers rather than grasping core concepts. The study examined whether automated MCQ generation using preference learning and ontology-based techniques could create more effective and meaningful questions. By leveraging fireflies-based preference learning, the system ranked text passages to extract key information for question stems, ensuring that the questions were informative and contextually relevant.

[3] "GENERATING AI LITERACY MCQS: A MULTI-AGENT LLM APPROACH": Jiayi Wang, Ruiwei Xiao, Ying-Jui Tseng, November 2024.

The AI Literacy MCQ Generation System in 2412.00970v1.pdf uses the Lang Graph framework with GPT-4o-mini for question generation. It employs multi-agent workflows including the Generator Agent, Language Critique Agent, and IWF Critique Agent for refining MCQs. The system aligns questions with Bloom's Taxonomy and uses rule-based and LLM-based validation to ensure quality.

[4] "DESIGN OF ONLINE QUIZ SYSTEM FOR PRACTICE & ASSESSMENT DURING COVID-19", Kapeel Dev, Rajesh Kumar, Imran saeed, Farooq Iqbal, Khalid Hussain, S. Madeeha Zaidi, M. Waqas Pasha6 in International Journal of Advanced Trends in Computer Science and Engineering, August 2021.

The Online Quiz System in ijatcse261042021.pdf is built using PHP, HTML, CSS, JavaScript, JQuery, and AJAX for frontend and interactivity. MySQL is used for database management, while Visual Studio serves as the development environment. The system follows an Iterative Model and supports CRUD operations for quiz management.

[5] "QUIZME: AN AI-POWERED QUIZ GENERATION SYSTEM", Prof. Abhishek Shivhare, Rohan Kumar Saini, Rishabh, Harshit Mahavar, Mukul Sharma in International Journal of Novel Reasearch and Development , April 2024.

The QuizMe: AI-Powered Quiz Generation System utilizes Next.js for building an interactive frontend, styled with Tailwind CSS and Shards UI Components for a responsive user experience. The backend is powered by Node.js with Express.js for API handling, while PostgreSQL is used as the database, managed efficiently through Prisma ORM. AI-powered quiz generation is enabled via the OpenAI API, ensuring contextually relevant questions. Additionally, JWT is used for authentication, and bcrypt secures user data by encrypting passwords.

[6] "AUTOMATIC MCQ GENERATION USING MACHINE LEARNING ALGORITHM" : Ganesh Wani, Kulashri Kale, Deep Kale , Shivam Shinde in International Research Journal of Engineering and Technology, June 2024.

The QuizMe system leverages Next.js for a dynamic and responsive frontend, styled with Tailwind CSS and Shards UI for an intuitive user experience. The backend is powered by Node.js with Express.js, ensuring efficient server-side logic and API handling. Data storage and

management are handled by PostgreSQL, with Prisma ORM simplifying database interactions.

[7] "MCQGEN: A LARGE LANGUAGE MODEL-DRIVEN MCQ GENERATOR FOR PERSONALIZED LEARNING": Ching Nam Hang, Pei-duo Yu, Chee Wei Tan in Institute of Electrical and Electronics Engineers, June 2024.

The MCQGen system utilizes GPT-4 combined with Retrieval-Augmented Generation (RAG) to generate high-quality MCQs by retrieving relevant external knowledge. It enhances question formation through Chain-of-Thought (CoT) reasoning and Self-Refine prompting, ensuring logical depth and accuracy. The system integrates an external knowledge base containing instructor-designed and student-created MCQs, improving contextual relevance. A crowdsourcing-based feedback mechanism refines generated questions by analyzing learner performance and educator input. This approach optimizes MCQ difficulty, engagement, and alignment with diverse educational needs.

[8] "GENMCQPRO: AI-DRIVEN MCQ REVOLUTION": Muralidharan A, Prem Raj, Rohan Vignesh S, Neyan AM, Dr. S. Hemamalini in Technix International Journal for Engineering Research, July 2024.

The GenMCQPro System leverages the T5 Transformer model to generate multiple-choice questions (MCQs) while ensuring contextual accuracy and diversity. For distractor generation, it employs sense2vec, which analyzes semantic similarity to create plausible yet challenging answer options. MultipartiteRank is used for extracting key concepts, while spaCy and NLTK handle text tokenization and preprocessing. The system runs on PyTorch and Transformers, integrating Levenshtein distance to measure text similarity and refine question quality. Built in a Python environment, it utilizes NumPy, pandas, and flashtext for data processing, ensuring efficiency in MCQ generation.

CHAPTER 3

PROBLEM STATEMENT

3. PROBLEM STATEMENT

The conventional quiz creation and evaluation process is often labor-intensive, time-consuming, and lacks adaptability to individual learning needs. To address these challenges, we propose the development of an AI-based quiz system that automates quiz generation, grading, and feedback. This system ensures personalized and adaptive assessments, enhancing accuracy, reducing educator workload, and fostering improved student engagement and learning outcomes.

CHAPTER 4

EXPERIMENTAL SETUP

4. EXPERIMENTAL SETUP

4.1 HARDWARE SETUP

- **Processor:** Intel Core i5 (or equivalent AMD Ryzen 5) and above for optimal performance.
- **RAM:** Minimum 8 GB of RAM to handle multiple queries and run the model smoothly.
- **Storage:** At least 256 GB SSD for faster read/write operations, or 500 GB HDD.
- **Network:** A stable internet connection (preferably 10 Mbps or higher) for handling queries and API interactions.
- **Operating System:** Windows 10 (64-bit)
- **Python Version:** Python 3.8 to ensure compatibility with modern Python libraries and frameworks.
- **Virtual Environment Setup:** Virtual environment setup (venv) for managing dependencies and avoiding conflicts with system-wide Python packages.

4.2 Software Setup

To efficiently run the "AI Based Quiz System" system, the following software tools and modules are required.

4.2.1 Frontend Technologies

1. **HTML (HyperText Markup Language)**
    - Provides the structure of the quiz system, such as login forms, quiz pages, and result displays.
2. **CSS (Cascading Style Sheets)**
    - Enhances the appearance of the website with styling, colors, fonts, and layouts.
    - You can use **CSS frameworks like Bootstrap** to make the design more responsive.
3. **JavaScript (JS)**
    - Adds interactivity to the quiz, such as timer functionality, dynamic content updates, and form validation.

4. **Jinja2 (Flask Templating Engine)**

- Used with Flask to generate dynamic HTML pages.
- Example: Displaying AI-generated quiz questions using Jinja2 loops.

5. **Bootstrap**

- A popular CSS framework that makes web pages mobile-friendly and visually appealing.
- Provides pre-designed buttons, forms, and layouts.

4.2.2 Backend Technologies

1. **Flask (Python Web Framework)**

- A lightweight framework for building web applications.
- Handles user authentication, form submissions, and API requests.

2. **Flask-MySQL (Database Integration)**

- Enables Flask to communicate with a MySQL database to store user data, quizzes, and responses.

3. **Session Management (Flask's Built-in Feature)**

- Keeps track of logged-in users and their quiz progress.
- Prevents unauthorized access to certain pages.

4. **MySQL (Relational Database)**

- Stores user credentials, quiz questions, AI-generated MCQs, and scores.
- Ensures efficient data retrieval and management.

5. **Google Gemini AI (google.generativeai)**

- AI model used to generate MCQ questions with options from a given paragraph or document.
- Extracts key information and converts it into quiz questions automatically.

6. **PyPDF2 (PDF Processing Library)**
    - Extracts text from PDF files for AI-based question generation.
    - Useful if users upload a document from which MCQs need to be created.
7. **Regular Expressions (re Library in Python)**
    - Helps in processing and extracting relevant content from the document.
    - Used for pattern matching (e.g., identifying headings, important terms).
8. **Datetime Module**
    - Manages timestamps for user activities (e.g., quiz start and end times).
    - Can be used to set quiz expiration times.

CHAPTER 5

PROPOSED SYSTEM & IMPLEMENTATION

5. PROPOSED SYSTEM & IMPLEMENTATION

5.1 BLOCK DIAGRAM OF PROPOSED SYSTEM

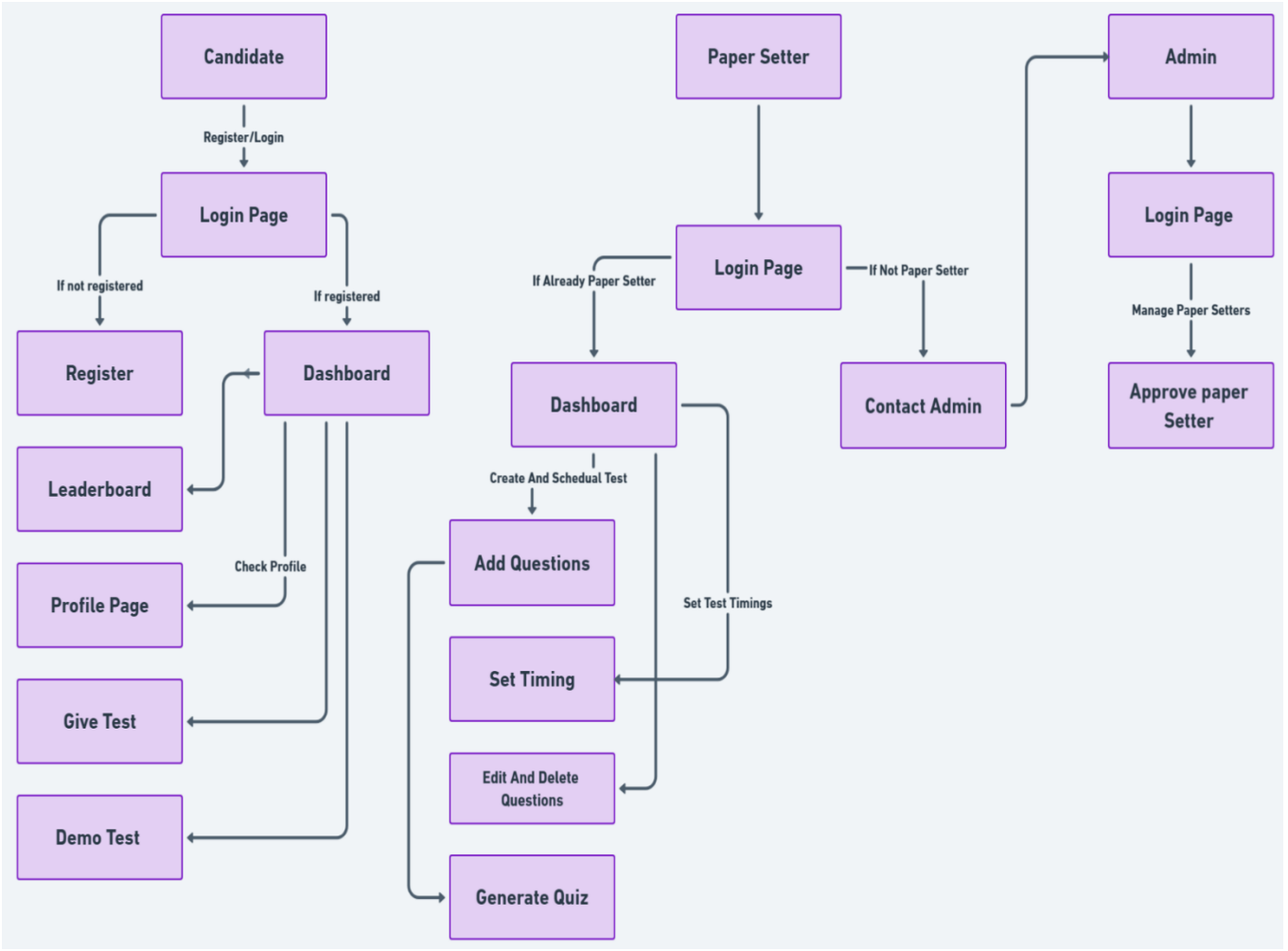


Fig: Block Diagram of AI Based Quiz System

5.2 DESCRIPTION OF BLOCK DIAGRAM

Candidate Workflow:

1. Start – A candidate initiates the process.
2. Login – The candidate attempts to log in.
    - If not registered, they must go through the Register process.
    - If registered, they proceed.
3. Actions after Login:
    - View My Profile.
    - Check Leaderboard.
    - Take the TEST (if available).
4. Test Process:
    - If a test is scheduled, the candidate can proceed.
    - If no test is scheduled, they must wait.
    - Once a test is completed, the results are updated and displayed.
5. Logout – The candidate can log out, leading to a stop.

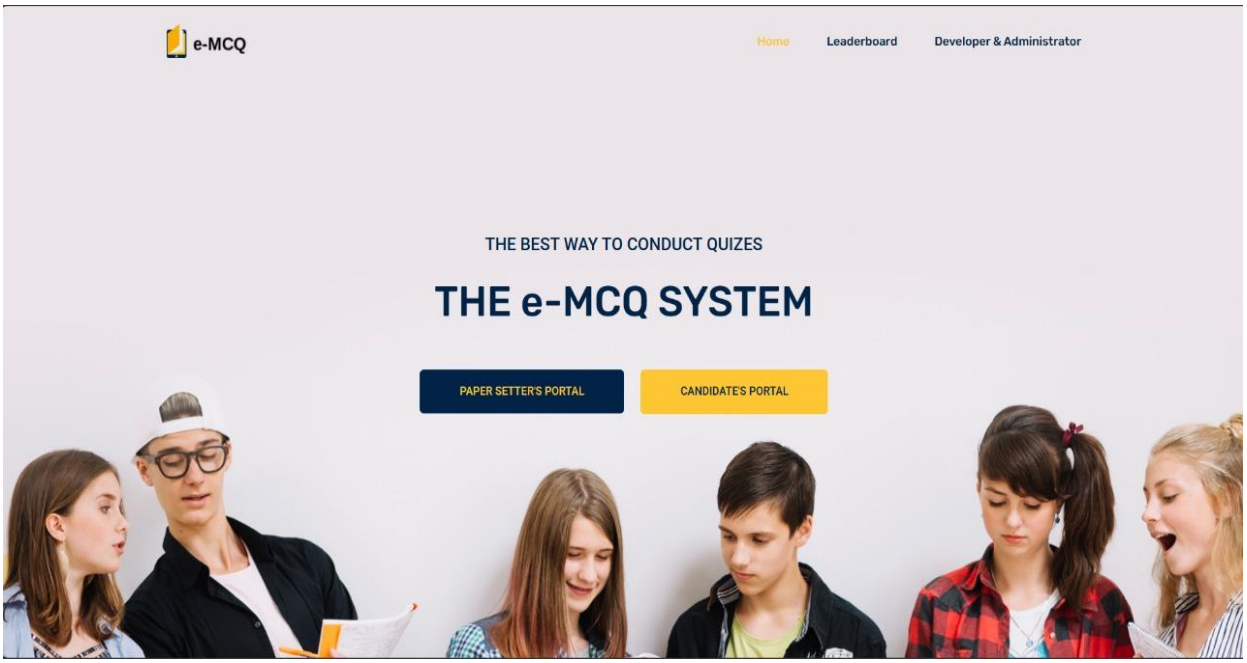
Paper Setter Workflow:

1. Login – The paper setter logs in.
2. Create Test – They can create a test and set its timings.
3. Logout – The paper setter can log out, leading to a stop.

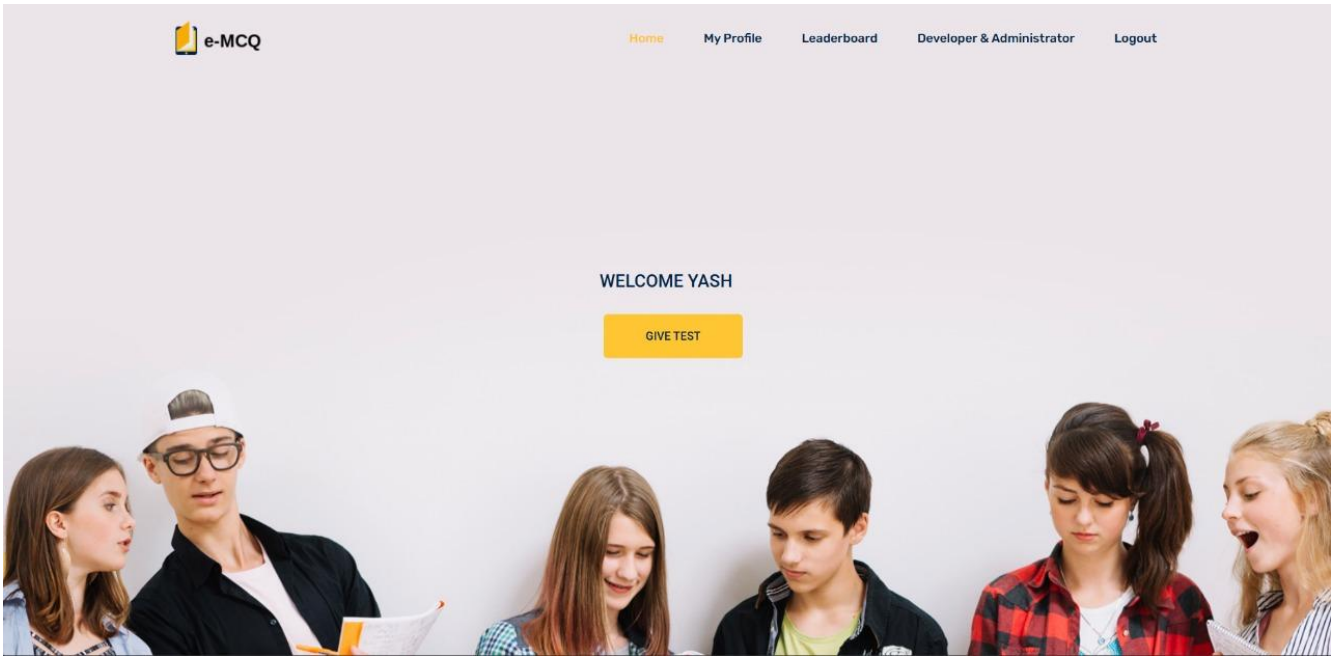
Decision Points:

- If the candidate is not registered, they must register before logging in.
- If no test is scheduled, the candidate has to wait.
- After completing the test, results are updated.

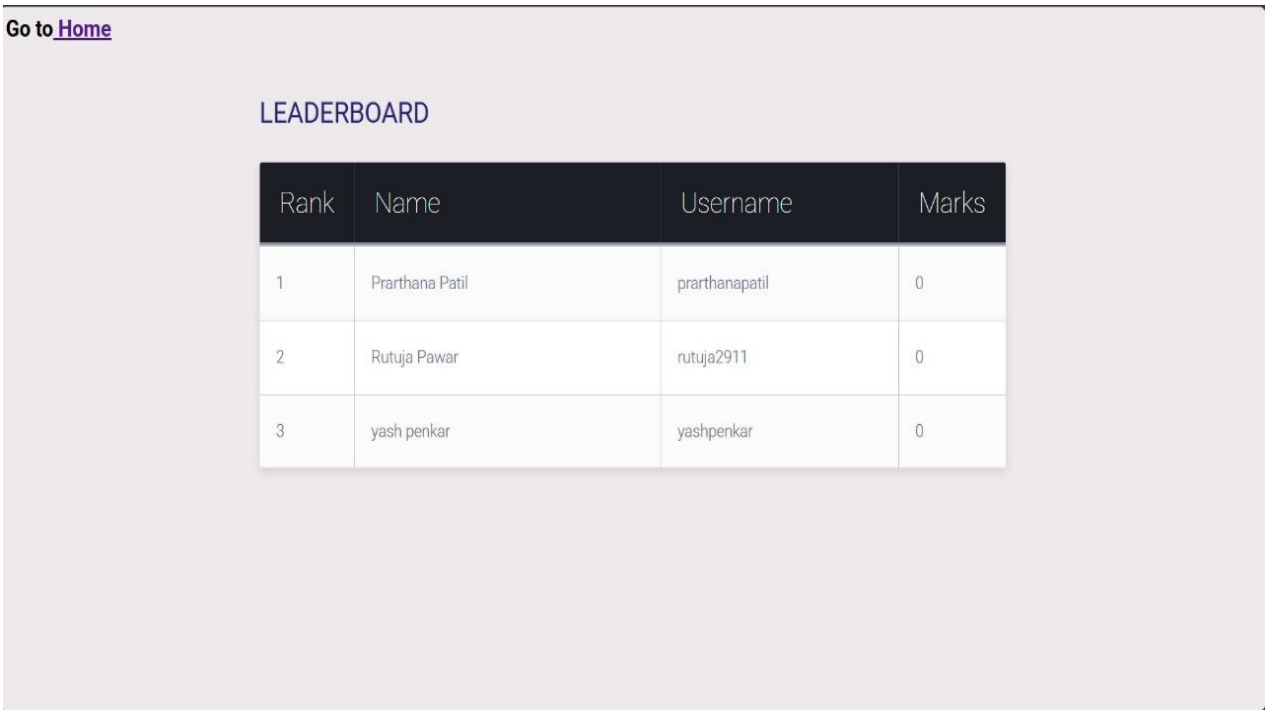
IMPLEMENTATION



Screenshot 1: Homepage



Screenshot 2: Candidate Portal



Screenshot 3: Leaderboard

Implementation of AI-Based Quiz System

This AI-powered quiz system is developed using **Flask**, **MySQL**, and **Google Gemini AI** to allow users to register, log in, attempt quizzes, and generate questions automatically using AI. The system consists of three main roles: **Admin, Paper Setter, and Candidate**, each with different functionalities.

Key Features and Functionalities

1. User Authentication and Roles

The system has role-based authentication where users can log in as **Admin, Paper Setter, or Candidate**.

- **Admin**: Approves paper setters and manages the quiz system.
- **Paper Setter**: Adds and manages questions.
- **Candidate**: Takes quizzes and views their scores.

Login & Registration

- Users can register by providing their details, including name, email, gender, DOB, username, and password.
- After successful registration, users are redirected to the login page.
- During login, credentials are checked against the **users** table in the database.
- Role-based redirection:
    - **Admin → Admin Panel**
    - **Paper Setter → Question Management Panel**
    - **Candidate → Home Page**

2. Quiz Management

The system supports manual question creation as well as **AI-generated questions** using **Google Gemini AI**.

Paper Setter Panel

- Paper setters can log in and add multiple-choice questions manually by entering:
    - Question Number
    - Question Text
    - Options (A, B, C, D)
    - Correct Answer
    - Marks
- The system saves this data in the **questions** table.

Editing & Deleting Questions

- Paper setters can view all existing questions.
- They can edit or delete questions from the **ps_view** panel.

3. AI-Based Question Generation

Instead of manually creating questions, paper setters can generate MCQs using AI.

How It Works:

1. The user uploads a **PDF document** containing educational content.
2. The system extracts text using **PyPDF2**.
3. The extracted text is sent to **Google Gemini AI** with a structured prompt.
4. AI generates **multiple-choice questions (MCQs)** with correct answers.
5. The generated questions are displayed and can be added to the database.

4. Quiz Attempting System

- Candidates can start a quiz after reading instructions.
- Questions are fetched one by one from the database.
- Candidates select an option and proceed to the **next question**.
- The system tracks answers and calculates marks in real time.
- Candidates can **navigate between questions** (Next & Previous).
- The final score is stored in the **leaderboard**.

5. Leaderboard and Results

- After quiz completion, candidates can view their scores.
- Scores are saved in the **leaderboard** table.
- The leaderboard displays rankings of all participants.

6. Admin Controls

- **Admin Login**: Only authorized admins can log in.
- **Approve Paper Setters**: Admins can approve or reject paper setter requests.
- **Manage Users**: Admins can view and manage users.
- **Reset Leaderboard**: Admins can clear all scores.

5.4 ADVANTAGES

- **Automated Quiz Generation –** AI can dynamically generate quizzes based on the provided content, reducing manual effort and saving time for educators.
- **Instant and Accurate Grading –** The system automatically evaluates responses, eliminating human errors and ensuring consistent and objective grading.
- **Personalized Learning Experience –** AI adapts quizzes based on a student's performance, providing customized assessments that cater to individual learning needs.
- **Reduced Educator Workload –** Automating quiz creation, grading, and feedback allows educators to focus on more critical aspects of teaching.
- **Enhanced Student Engagement –** Interactive and adaptive assessments make learning more engaging and motivate students to participate actively.
- **Scalability –** The system can handle a large number of students simultaneously, making it suitable for educational institutions, corporate training, and online learning platforms.
- **Data-Driven Insights –** AI can analyze quiz results and provide insights into student performance, helping educators identify knowledge gaps and adjust their teaching strategies.
- **Accessibility and Flexibility –** Students can take quizzes anytime, anywhere, supporting remote learning and self-paced study.

CHAPTER 6

CONCLUSION

6. CONCLUSION

The system streamlines user authentication and session management, ensuring secure and seamless access for users. It efficiently handles quiz creation, editing, and deletion, making it easy for educators and administrators to manage assessments. Real-time scoring and instant feedback enhance the user experience, providing learners with immediate insights into their performance. Additionally, the platform supports dynamic leaderboards, fostering a competitive learning environment that motivates users to improve. Designed to be both secure and scalable, the system is an ideal solution for online assessments and learning, ensuring reliability and efficiency in the digital education landscape.

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